SESSION 4

CQs:

1. CQ29: If a newly instantiated RoadSign is asserted to have the hasShape property linking it to the shape_triangle instance and hasBorderColour linking it to color_red, will the reasoner automatically infer that it belongs to the DangerWarningSign class?
2. CQ30: Can a sign be automatically classified as a ProhibitorySign if it has a shape_circle and a color_red border, even if the user forgets to manually assign it to that specific subclass?
3. CQ31: What class is automatically inferred for an unknown sign with shape_circle and color_blue ground colour?
4. CQ32: If the Disjointness constraints are removed between main sign categories, will the ontology reasoner successfully classify overlapping instances without throwing inconsistency errors?

LLM: Gemini

LLM Prompt: Role: You are an expert Ontology Engineer specializing in traffic regulations and knowledge representation.

Objective: Generate Manchester Syntax definitions for the main road-sign classes in order to enable automatic classification by a reasoner. Provide Necessary and Sufficient conditions (Equivalent To). Identify and resolve logical inconsistencies related to property characteristics (e.g., removing Disjoint constraints to support the Open World Assumption).

Scope & Domain: Domain: Section A, C, and D of the Vienna Convention on Road Signs and Signals. Target Namespace: http://www.ontology.org/traffic/vienna-convention#.

Key Entities: The class definitions to generate are for DangerWarningSign, ProhibitorySign, and MandatorySign.

Boundaries: DO NOT generate instances or object properties in this step. ONLY generate the Manchester Syntax strings for the equivalent class axioms.

Competency Questions: The ontology must be able to answer the following questions:

- If a newly instantiated RoadSign is asserted to have the hasShape property linking it to the shape_triangle instance and hasBorderColour linking it to color_red, will the reasoner automatically infer that it belongs to the DangerWarningSign class?

- Can a sign be automatically classified as a ProhibitorySign if it has a shape_circle and a color_red border?

Structural Requirements: Provide the equivalent class axioms using vienna-convention1:RoadSign as the base to avoid disjointness conflicts from previous sessions. Use vienna-convention1:hasShape, vienna-convention1:hasBorderColour, and vienna-convention1:hasGroundColour properties. Remove Functional characteristics from colors to allow real-world dataset exceptions.

Output Format: Provide ONLY the raw Manchester Syntax strings for the class definitions. Do not write explanations.